

Contestant Number: _____

Time: _____

Rank: _____



LINUX OPERATING SYSTEM FUNDAMENTALS (350)

STATE 2021

TOTAL POINTS

_____ (100 points)

Test Time: 60 minutes

Please select the answer that best answers the question.

- 1) What utility can be used to create partitions on a disk in Ubuntu?
 - a. fdisk
 - b. parted
 - c. cgdisk
 - d. all of the above
- 2) If you run the command: “mkfs /dev/sda2” what filesystem, would be applied to the disk partition?
 - a. Ext2
 - b. Ext3
 - c. Ext4
 - d. Xfs
- 3) Which of the following will *not* list the disks partitions?
 - a. fdisk -l /dev/sda
 - b. gdisk -l /dev/sda
 - c. cgdisk -l /dev/sda
 - d. parted -l /dev/sda
- 4) From the output below, identify which is a hidden file.


```
drwxrwxr-x 2 ryan ryan 4.0K Aug 14 19:57 test
drwxrwxr-x 2 ryan ryan 4.0K Aug 14 20:06 .test1
-rw-rw-r-- 1 ryan ryan 0 Aug 14 19:57 .test2
-rwxrwxr-t 1 ryan ryan 0 Aug 14 19:58 test3
```

 - a. test
 - b. test1
 - c. test2
 - d. test3
- 5) Which command will *not* change the hostname to **BPA1** in Ubuntu?
 - a. sudo hostname BPA1
 - b. sudo hostnamectl set-hostname BPA1
 - c. sudo echo BPA1 > /etc/hostname
 - d. sudo hostname --set BPA1
- 6) Which ls argument will list all hidden files and folders?
 - a. -t
 - b. -h
 - c. -a
 - d. -i

- 7) Which chmod commands will make the file runme.sh executable?
- a. chmod +X ./runme.sh
 - b. chmod +x ./runme.sh
 - c. chmod 667 ./runme.sh
 - d. chmod 666 ./runme.sh
- 8) The command “**sudo mount /dev/sdb1 /Project**” will successfully mount the first partition of the second disk to the **/Project** directory.
- a. True
 - b. False
- 9) Which command will successfully list all logical volumes in the volume group named Project?
- a. lvm Project
 - b. lvs Project
 - c. pvs Project
 - d. vgs -l Project
- 10) There is an LVM Volume group named Project on this system, and you have been tasked with create a new logical volume in that volume group named Home, which needs to be 2GB in size. Which command will successfully complete that task?
- a. sudo lvcreate -n Home Project --size=2GB
 - b. sudo lvcreate -n Home --vg Project --size=2GB
 - c. sudo lvcreate -n Home Project -L2GB
 - d. sudo lvcreate -n Home --vg Project -L2GB
- 11) Which command will display the ip address of all network adapters on the system?
- a. ip link
 - b. ip a
 - c. ip --all
 - d. ip show all
- 12) Which command will successfully change the IP Address of the Network Adapter ens192 to 10.10.0.5 with subnet of 255.255.255.0?
- a. sudo nmcli connection modify ens192 ipv4.addresses 10.10.0.5/255.255.255.0
 - b. sudo nmcli connection modify ens192 ipv4.addresses 10.10.0.5/24
 - c. sudo nmcli connection add ens192 ipv4.addresses 10.10.0.5/255.255.255.0
 - d. sudo nmcli connection add ens192 ipv4.addresses 10.10.0.5/24
- 13) An LVM partition has been created on disk sdb for use with LVM. What command would *correctly* create a volume group named Project using that partition in Ubuntu?
- a. sudo vgcreate -n Project -d /dev/sdb1
 - b. sudo vgcreate Project /dev/sdb1
 - c. sudo vgcreate -n Project -p /dev/sdb1
 - d. sudo vgcreate Project sdb1

- 14) Which utility *cannot* be used to transfer files directly to another server on the network?
- scp
 - rsync
 - tar
 - ftp
- 15) Which of the following is *not* a valid firewall that can be used in Linux?
- Iptables
 - Firewalld
 - Ufw
 - Selinux
- 16) Which command would *correctly* set the NIC ens192 to use static assigned IP instead of DHCP?
- sudo nmcli connection edit ens192 ipv4.method manual
 - sudo nmcli connection modify ens192 ipv4.method static
 - sudo nmcli connection edit ens192 ipv4.method static
 - sudo nmcli connection modify ens192 ipv4.method manual
- 17) Which command would *correctly* change the UID of a user named Tom to 1004?
- sudo usermod -U 1004 Tom
 - sudo usermod -u 1004 Tom
 - sudo useradd -u 1004 Tom
 - sudo useradd -U 1004 Tom
- 18) What command would *correctly* create a new group called Admins with a GID of 1005?
- sudo groupadd -G 1005 Admins
 - sudo groupmod -g 1005 Admins
 - sudo groupadd -g 1005 Admins
 - sudo groupmod -G 1005 Admins
- 19) On the system is a user named Billy, and he needs to be added to the two groups, Admins and Managers without affecting his primary group. Which command would successfully accomplish this?
- sudo usermod -aG Admins,Managers Billy
 - sudo usermod -g Admins,Managers Billy
 - sudo usermod -ag Admins,Managers Billy
 - sudo usermod -G Admins,Managers Billy
- 20) You need to add a new user named Jane, but they need their home folder to be /mnt/Users/Jane instead of in the default /home directory. Which command would successfully accomplish this?
- sudo useradd -h /mnt/Users/Jane Jane
 - sudo useradd -m /mnt/Users/Jane Jane
 - sudo useradd -d /mnt/Users/Jane Jane
 - sudo useradd -s /mnt/Users/Jane Jane

- 21) Which command would be used to initialize a disk or partition for use with LVM volume groups?
- a. pvs
 - b. pvcreate
 - c. pvchange
 - d. pvformat
- 22) Which command will list all disk partitions and their filesystem type?
- a. lsblk
 - b. blkid
 - c. df
 - d. fdisk
- 23) In order for new user to run a command with root privileges in Ubuntu, they must first be added to which group?
- a. administrator
 - b. admin
 - c. sudo
 - d. root
- 24) What command will update all available packages in Ubuntu?
- a. sudo apt update
 - b. sudo dpkg --update
 - c. sudo apt upgrade
 - d. sudo yum upgrade
- 25) When installing Ubuntu Server OS, which GUI Desktop would be installed by default?
- a. Gnome
 - b. KDE
 - c. Unity
 - d. None
- 26) Which of the commands would *not* be a valid way to search the Ubuntu repositories for a package named vim?
- a. apt-cache search vim
 - b. apt search vim
 - c. apt-get search vim
 - d. aptitude search vim
- 27) Which command can be used to upgrade the currently installed Ubuntu 18.04 to the latest Ubuntu 20.04 version?
- a. sudo apt dist-upgrade
 - b. sudo apt upgrade
 - c. sudo do-release-upgrade
 - d. sudo apt full-upgrade

28) You just installed the OpenSSH Server package on the system, and now you need to start it and set it to run at boot. Which single command below would complete both tasks?

- a. `sudo systemctl start sshd`
- b. `sudo systemctl enable sshd`
- c. `sudo systemctl enable --now sshd`
- d. `sudo systemctl is-enabled --true sshd`

29) Which command would stop the httpd web service from starting on the next boot?

- a. `sudo systemctl stop httpd`
- b. `sudo systemctl disable httpd`
- c. `sudo systemctl stop --disable httpd`
- d. `sudo systemct is-disabled --now httpd`

30) Using the journalctl utility to view system logs, what command would allow you to filter the logs for anything referencing the ssh service?

- a. `sudo journalctl --search ssh`
- b. `sudo journalctl -u ssh`
- c. `sudo journalctl -t ssh`
- d. `sudo journalctl -S ssh`

31) Which command will only display log entries that are at alert and critical level?

- a. `sudo journalctl -s crit`
- b. `sudo journalctl -f alert`
- c. `sudo journalctl -p alert`
- d. `sudo journalctl -b crit`

32) Looking at the attributes below for the file labeled “file1”, you can conclude that anyone in the admins group would have both read and write access to the file.

-rw-r--rw- 1 ryan admins 0 Aug 28 18:08 file1

- a. True
- b. False

33) Looking at the attributes below for the file labeled “file2”, which user would have both read and write access to the file?

-r-xrw---x 1 ryan zack 0 Aug 28 18:11 file2

- a. ryan
- b. zack
- c. Everyone
- d. Neither

- 34) You have a user that says they can't open up a folder called /mnt and get permission denied when trying to run `cd /mnt`. You have already confirmed their group has been assigned to the folder. What directory permission do they need added for their group in order to open the folder?
- Write
 - Read
 - Execute
 - All of the above
- 35) If you were to change permissions to a file using the command: "`chmod 740 filename`", what would the resulting permissions look like when viewing the file?
- `-rwxr--r--`
 - `-rwxr-----`
 - `-rw--wxr--`
 - `----r--rwx`
- 36) Review the permissions below for the folder Database. The user Tom has been added to the group Management but has no write access while the user Adam, who is not part of the Management group, has full read and write access. Which command below would remove write access for other users and provide Tom in the Management group full read/write access?
- drwxr-xrwx 2 root Management 4.0K Aug 28 18:34 Database**
- `sudo chown :Tom Database`
 - `sudo chmod 757 Database`
 - `sudo chmod 775 Database`
 - `sudo chown Tom:Tom Database`
- 37) If you were to set permissions of **546** on a script called `runme.sh`, users in which category would be given the ability to execute the script?
- Owner
 - Group
 - Others
 - None of the above
- 38) You need to apply permissions to a folder for three different groups. Which command would allow you to accomplish this?
- `chown`
 - `chmod`
 - `setfacl`
 - Attribute

- 39) Which command would successfully filter out any mention of errors from the `/var/log/syslog` file and output them to a file called `/var/log/errors.log` without overwriting any existing data?
- a. `grep -i "error" /var/log/syslog > /var/log/errors.log`
 - b. `find /var/log/syslog -name "error" >> /var/log/errors.log`
 - c. `grep -i "error" /var/log/syslog >> /var/log/errors.log`
 - d. `find /var/log/syslog -name "error" > /var/log/errors.log`
- 40) What port does the SSH service use by default?
- a. 23
 - b. 22
 - c. 25
 - d. 21
- 41) You have set up a secure webserver and opened up port 80 on the local firewall, yet clients are unable to access the web server's https addresses. Which additional firewall port needs to be opened?
- a. 8080
 - b. 445
 - c. 443
 - d. 465
- 42) When viewing processes with the `top` utility, which key would sort the list by CPU usage?
- a. T
 - b. C
 - c. P
 - d. M
- 43) Which command below would *not* show all processes running by the user Adam?
- a. `top -u adam`
 - b. `ps -u adam`
 - c. `pstree adam`
 - d. `tasklist adam`
- 44) You need to add a new disk partition, ***sdc1***, into your existing volume group named ***Database***. Which command will accomplish this?
- a. `vgreszie Database /dev/sdc1`
 - b. `vgetend Database /dev/sdc1`
 - c. `vgexpand Database /dev/sdc1`
 - d. `vgadd Database /dev/sdc1`

- 45) All data has been migrated off of disk **sdb1** and is now ready to be removed from the volume group **Database**. What command will successfully complete the removal?
- a. `vgremove Database /dev/sdb1`
 - b. `vgreduce Database /dev/sdb1`
 - c. `pvremove Database /dev/sdb1`
 - d. `pvmove Database /dev/sdb1`
- 46) The command **pvmigrate** is used to migrate data off of a physical disk before it can be removed from a volume group.
- a. True
 - b. False
- 47) Which network utility will allow you to set a NIC to start at boot?
- a. `nmcli`
 - b. `ip`
 - c. `ifconfig`
 - d. `ipconfig`
- 48) After making changes to the grub config file, what command do you need to run to apply changes?
- a. `grub-install`
 - b. `update-grub`
 - c. `grub-mkconfig`
 - d. `save-grup`
- 49) In Ubuntu selinux is enabled by default.
- a. True
 - b. False
- 50) The samba service provides file and print services for both Linux and Windows client systems.
- a. True
 - b. False